

The Ontological Neutrality Theorem: Structural Constraints for Neutral Substrates Under Persistent Interpretive Disagreement

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Abstract

Modern data systems must support stable shared reference across persistent interpretive disagreement. This requirement imposes structural constraints on any ontology intended to function as a neutral substrate. This paper establishes a necessary and sufficient condition for ontological neutrality under an explicitly stated contestability assumption and scope condition. The central result is a biconditional: a substrate ontology satisfies neutrality if and only if its foundational layer is restricted to framework-invariant referential structure and does not assert causal or normative conclusions as substrate-layer facts. Neutrality here is the conjunction of two requirements: interpretive non-commitment and extension stability. The result holds under a contestability assumption for causal and normative propositions, and under the scope condition that the substrate's framework-invariant referential structure, including its referential regime, is fixed at the substrate layer and is not itself contested across the admissible frameworks under consideration. A referential regime consists of the individuation, co-reference, and persistence conditions by which entities, occurrences, and institutional artifacts are fixed and tracked across time, records, jurisdictions, or interpretive contexts. The result does not require causal or normative vocabulary to be absent from the substrate. Rather, it

requires that causal and normative claims be represented, if at all, as reified, attributed, provenance-bearing assertions whose truth is evaluated within interpretive frameworks rather than asserted by the substrate itself. The paper does not propose a specific ontology or protocol. It identifies the structural constraints that any ontology must satisfy to function as a neutral substrate across admissible interpretive frameworks under persistent disagreement.

Keywords: Formal ontology; ontological neutrality; neutral substrates; accountable records; referential regimes; interpretive non-commitment; extension stability; persistent interpretive disagreement; reification; provenance; pre-causal; pre-normative

Preview of Main Result. *Let $\mathcal{S}$ be a substrate ontology intended to function as a neutral substrate across admissible interpretive frameworks, and let $\mathbb{F}$ be the set of all admissible frameworks. Suppose the domain satisfies the contestability assumption for causal and normative propositions and the scope condition that the framework-invariant referential structure asserted by $\mathcal{S}$, including its referential regime, is fixed at the substrate layer and is not itself contested across the admissible frameworks under consideration. Then $\mathcal{S}$ satisfies the neutrality requirements of interpretive non-commitment and extension stability if and only if its foundational layer is restricted to framework-invariant referential structure and does not assert causal or normative conclusions as substrate-layer facts.*

Informally. The result applies to systems that adopt the design objective of maintaining a neutral substrate under persistent interpretive disagreement. Under this objective, the theorem establishes that a neutral substrate must be pre-causal and pre-normative at the level of assertion. This does not mean that causal or normative concepts are absent, unimportant, or unreal. It means that causal and normative conclusions may not be asserted as substrate-layer facts. They may instead be represented as reified, attributed, provenance-bearing claims whose truth is evaluated within interpretive frameworks layered above the substrate.

The substrate may assert framework-invariant referential structure: entities, occurrences, institutional artifacts, provenance, the referential regimes by which relevant referents are individuated, co-referred, and tracked across time or context, and other structural relations that assert neither causal efficacy nor normative force. If

admissible frameworks disagree about any part of that referential structure, including the regimes, then neutrality in the sense defined here is unavailable until that contestation is resolved or scoped out. Systems that enforce interpretive convergence through governance or shared authority, or that do not adopt the neutral-substrate design objective, are not subject to these constraints.

The formal definitions, assumptions, and proof appear in Sections 3 and 4.

1 Introduction: Accountability, Disagreement, and Neutral Substrates

Modern data systems must support stable shared reference across persistent legal, political, analytic, causal, and normative disagreement. Legal interpretations diverge across jurisdictions and over time (Dworkin 1986; Hart 1961; Berman 2007); political actors contest responsibility and intent (Bovens 2007; Hood 2011); analytic frameworks disagree about causation and attribution (Woodward 2003). These disagreements are not anomalies to be resolved prior to data representation. They are structural features of institutional environments that data systems must accommodate while maintaining stable reference to shared entities, occurrences, and institutional artifacts.

Accountability is one motivating setting in which this requirement becomes especially visible. Accountability requires more than the collection of records or the publication of datasets (Fox 2007; Bovens et al. 2014). It requires the ability to refer stably to entities, occurrences, and institutional artifacts (Gruber 1993; Smith and Ceusters 2010) while allowing competing explanations, evaluations, and judgments to coexist. A system that collapses disagreement by embedding a single causal or normative interpretation into its foundational representation restricts accountability in settings characterized by persistent interpretive disagreement, because it forecloses contestation, comparison, and framework-relative evaluation (Bowker and Star 1999; Winner 1980).

This paper adopts a different framing of the design challenge. Rather than enforcing interpretive agreement, or treating disagreement primarily as semantic heterogeneity to be resolved through alignment, the goal is to identify the structural constraints under which an ontology can function as a neutral substrate: a shared representational

base that maintains stable reference across incompatible interpretive frameworks.

Design Problems in Shared Representation

Ontology design has traditionally addressed two related but distinct representation problems. First, systems may assume *shared interpretation*. In this setting, participants agree on a common ontology that provides a unified interpretation of shared representations. Second, systems may face *semantic heterogeneity*. Here multiple ontologies exist, but disagreement is treated as a translation problem addressed through alignment, mapping, or mediation mechanisms.

The problem considered in this paper is different. Many institutional and accountability-oriented environments exhibit *persistent interpretive disagreement*. An *interpretive framework* is a coherent set of rules, principles, assumptions, methods, or norms used to evaluate, explain, classify, or assign meaning to shared referents. Legal, political, causal, analytic, and normative frameworks are examples of interpretive frameworks in this sense. Such frameworks may reach incompatible conclusions about the same entities, occurrences, or institutional artifacts while continuing to operate over a shared representational base.

Table 1: Ontology Design Strategies Across Distinct Semantic Conditions

Setting	Ontology Structure	Mechanism
Shared interpretation	Shared ontology	Unified interpretation
Semantic heterogeneity	Multiple ontologies	Alignment / Mediation
Persistent disagreement	Neutral substrate	Multiple frameworks

Traditional ontology frameworks regulate how agreement is structured across systems. Neutral substrate architectures instead regulate how disagreement can coexist without forcing revision of the shared representational base.

This paper examines the implications of this requirement for ontology design. Specifically, it addresses the conditions under which an ontology can function as a *neutral substrate* for accountability: a shared representational layer that supports stable reference to entities, occurrences, and institutional artifacts while allowing multiple incompatible interpretive frameworks to coexist. Under this conception of neutrality, the substrate must remain usable across divergent interpretive frameworks without requiring revision of the substrate itself. A neutral substrate, in this sense,

is a foundational representation that supports such coexistence without embedding the causal or normative conclusions on which those frameworks may disagree.

In this paper, a *substrate ontology* is an ontology that serves as the shared representational base across interpretive frameworks. Its foundational layer provides entities, occurrences, institutional artifacts, provenance structures, and the stable referential structure over which interpretive frameworks reason and derive conclusions. Section 3 formalizes this stable referential structure as a *referential regime*: the substrate-level basis for individuating referents, determining co-reference, and tracking persistence across time, records, jurisdictions, or interpretive contexts. For brevity, this ontology may be referred to as the *substrate*.

This requires a careful selection of the Level of Abstraction (LoA) (Floridi 2008) at which the substrate operates. Following Floridi, an LoA is defined by the set of observables it makes available for representation. In an ontological substrate, these observables correspond to the predicate symbols available in the foundational vocabulary. For a substrate to remain neutral, its foundational assertions must provide stable shared reference without settling the causal or normative questions on which admissible interpretive frameworks may disagree. Section 3 states this assertion-level constraint formally.

Conceptually, the architecture considered here separates three roles. The substrate provides shared referents and framework-invariant referential structure. Interpretive frameworks reason over that structure using their own rules, assumptions, methods, or norms. The conclusions produced by those frameworks may disagree without requiring revision of the substrate. Section 3 formalizes these roles.

The key design question is therefore what the substrate may assert while remaining usable across incompatible interpretive frameworks.

This form of neutrality imposes strict design constraints. Causal and normative conclusions may be derived by interpretive frameworks, and causal or normative claims may be represented as attributed content. The substrate itself, however, must not settle those conclusions as its own. The formal requirements that make this constraint precise are introduced in Section 3.

This paper adopts the following design constraint. Under persistent disagreement, extensions to the substrate may introduce incompatible causal accounts or normative attributions, and those extensions are not required to converge. Substrate revision is treated as an impermissible mechanism for resolving interpretive conflict: the

substrate must remain stable across extensions whose conclusions are mutually inconsistent. This is a deliberate design choice, not a universal requirement. Systems that can enforce interpretive convergence through coordinated governance or shared authority do not face this constraint; the results of this paper apply only to substrates explicitly designed to remain stable without such coordination.

The contribution of this paper is not the observation that interpretive frameworks may disagree about causal or normative conclusions. That fact is widely recognized across legal theory, political analysis, and causal modeling. The contribution is to show that, in settings where substrate revision is treated as an impermissible mechanism for resolving interpretive conflict, this disagreement imposes structural constraints on ontology design, and to establish those constraints formally.

The central result is a biconditional: under the stated contestability assumption and referential-invariance scope condition, a substrate ontology satisfies neutrality if and only if its foundational layer is restricted to framework-invariant referential structure and does not assert causal or normative conclusions as substrate-layer facts.

Neutrality is therefore possible, under the stated assumptions and scope conditions, if and only if the substrate is pre-causal and pre-normative. Pre-causal and pre-normative do not mean acausal, norm-denying, or unable to represent causal or normative discourse. They mean that the substrate makes no causal or normative commitments. Such propositions may be represented as reified, attributed, provenance-bearing content or introduced within interpretive frameworks layered above the substrate. The precise conditions under which extensions may be introduced are stated in Section 3.

The remainder of the paper proceeds as follows. Section 2 reviews related work in ontology design, legal informatics, and causal modeling, situating the present contribution. Section 3 defines the notion of a neutral substrate, the referential-invariance scope condition, and the two neutrality requirements. Section 4 establishes the central result: a neutral substrate satisfies both requirements if and only if its foundational layer is restricted to framework-invariant referential structure and does not assert causal or normative conclusions as substrate-layer facts. Section 5 discusses the implications of this result for ontology design, clarifying the substrate-level exclusions and requirements, and what may be externalized to interpretive frameworks. Section 6 summarizes the resulting constraints and discusses their relevance for ontology design in accountability contexts.

2 Related Work

The constraints developed in this paper intersect with several lines of prior work: upper ontologies in formal ontology research, domain ontologies for legal and institutional reasoning, and causal modeling frameworks. This section situates the present contribution relative to each, clarifying both debts and departures.

2.1 Political Philosophy: Reasonable Pluralism

The premise that disagreement is structural rather than resolvable draws on Rawls’ concept of reasonable pluralism (Rawls 1993). Rawls argues that under conditions of free inquiry, reasonable persons will arrive at incompatible comprehensive doctrines.

This paper transposes that insight to ontology design: if interpretive disagreement is a permanent feature of accountability contexts, then neutral substrates must accommodate it structurally rather than resolving it by fiat. The central result formalizes this accommodation as a structural constraint on substrate-layer commitments. Rawls’ argument is not imported as a normative foundation, but as an analogy for the persistence of disagreement that motivates the need for structural accommodation.

2.2 Upper Ontologies: BFO and DOLCE

The Basic Formal Ontology (BFO) (International Organization for Standardization and International Electrotechnical Commission 2021; Arp et al. 2015) and the Descriptive Ontology for Linguistic and Cognitive Engineering (DOLCE) (Borgo et al. 2022a; Masolo et al. 2003) represent two influential approaches to foundational ontology design. Both aim to provide domain-independent upper-level categories that can be extended for specific applications. A recent special issue (Borgo et al. 2022b) provides systematic comparison of seven foundational ontologies (BFO, DOLCE, GFO, GUM, TUpper, UFO, YAMATO) through common modeling cases, illustrating both their shared commitments and divergent design choices.

BFO distinguishes continuants from occurrents and provides a realist framework grounded in scientific practice. DOLCE takes a more cognitive and linguistic orientation, emphasizing the role of conceptualization in ontological commitment.

Both have been widely adopted in biomedical, engineering, and information systems contexts.

The present work addresses a different question: it analyzes the logical consequences of embedding interpretive commitments at the substrate layer, rather than the adequacy or intent of any particular foundational ontology. BFO and DOLCE provide categorical structure for what exists; they do not explicitly address the conditions under which an ontology can remain neutral across interpretive disagreement. These frameworks may be extended in practice to include causal and normative relations as first-class ontological commitments. The result established here implies that such extensions, if embedded at the substrate layer, compromise neutrality in the sense required for accountability systems.

In Floridi’s terms, while BFO and DOLCE provide foundational ontological structure, they do not mandate a specific Level of Abstraction that excludes interpretive predicates. Our contribution is to show that for the specific purpose of neutrality, such a mandate is a logical necessity.

Guarino’s analysis of ontological commitment (Guarino et al. 2009) clarifies what it means for an ontology to assert something; the present contribution identifies which such assertions are incompatible with neutrality.

This paper’s constraints are therefore compatible with BFO or DOLCE as upper-level scaffolding, provided that causal and normative commitments are externalized to interpretive layers rather than embedded in the foundational substrate. Recent work on formal alignment between BFO and DOLCE (Masolo et al. 2025) demonstrates that even foundational ontologies with divergent commitments can be mapped systematically, reinforcing the feasibility of neutral substrates that support multiple interpretive frameworks.

Whether specific versions or extensions of BFO, DOLCE, or other upper ontologies satisfy these constraints in practice is an empirical question beyond the scope of this paper. Such compatibility assessments require detailed analysis of particular ontology versions and their commitments, work that would benefit from collaboration with the communities that maintain and extend these frameworks.

2.3 Legal Ontologies: LKIF and Related Approaches

Legal ontologies, e.g., the Legal Knowledge Interchange Format (LKIF) (Hoekstra et al. 2007), and related frameworks (Sartor et al. 2011; Breuker and Hoekstra 2009) aim to represent legal concepts, norms, and reasoning patterns in machine-processable form. These ontologies typically include deontic categories, such as obligation, permission, and prohibition, as core primitives, along with relations for legal causation, responsibility, and compliance.

Such frameworks are valuable for legal reasoning and case analysis within a fixed normative framework. However, they are not designed to function as neutral substrates across incompatible legal or political interpretations. By embedding deontic conclusions as ontological facts at the foundational layer, such designs commit the ontology to a particular normative stance. When legal interpretations diverge, as they routinely do across jurisdictions, over time, or under political contestation, the substrate ontology itself must be revised to accommodate disagreement.

The result established here applies directly to such designs. LKIF and similar ontologies are appropriate for interpretive layers built atop a neutral substrate, but they cannot themselves serve as the substrate if neutrality across disagreement is required.

2.4 Causal Modeling: Pearl and Structural Approaches

The structural causal modeling framework developed by Pearl (Pearl 2009) and extended by others (Woodward 2003; Spirtes et al. 2000) provides rigorous tools for representing and reasoning about causation. The do-calculus and related interventionist semantics have become standard in causal inference across multiple disciplines.

These frameworks make explicit causal commitments: they assert that certain variables cause others under specified structural assumptions. Such commitments are essential for causal reasoning but are inherently model-dependent. Different causal models, each internally consistent, may disagree about which relations are causal, which variables are confounders, and which interventions produce which effects.

The present work does not dispute the value of causal modeling for explanation

and decision-making. Rather, it observes that causal conclusions are framework-dependent in the same structural sense as normative conclusions, as formalized subsequently in Lemma 4.1. When treated as substrate commitments, causal relations commit the ontology to a particular causal model. For neutral substrates, causal reasoning is therefore located in interpretive layers where model assumptions are explicit and disagreement can be represented without substrate revision.

2.5 Philosophy of Information: Levels of Abstraction

The methodological framework of this paper is informed by Floridi’s Philosophy of Information (Floridi 2008 2011). Floridi introduces the Method of Levels of Abstraction (LoA) as a way to specify the epistemic range of a system through a set of observables. In ontological terms, these observables correspond to the predicate symbols available in the system’s representational vocabulary. Crucially, Floridi distinguishes between the LoA (the lens through which a system is viewed) and the Level of Organization (LoO), which refers to the internal structural hierarchy of the system itself.

While Floridi uses these concepts to analyze the nature of data and knowledge, this paper applies them specifically to the problem of ontological neutrality. Under this view, neutrality is a function of operating at a foundational LoO with a deliberately restricted LoA. This paper extends Floridi’s work by proving that causal and normative predicates cannot be asserted at the substrate layer without violating neutrality under extension by admissible interpretive frameworks.

2.6 Information Infrastructure: Boundary Objects

Star and Griesemer’s concept of boundary objects (Star 1989; Star and Ruhleder 1996) and subsequent work on classification systems (Bowker and Star 1999) address how representations can be shared across communities with divergent interpretive frameworks. Boundary objects are plastic enough to adapt to local needs yet robust enough to maintain identity across sites.

The neutral substrate concept developed here can be understood as a formalization of this insight: the substrate provides the shared identity structure, while interpretive frameworks supply local meaning. The theorem specifies what cannot be shared at the boundary level without losing plasticity.

2.7 Positioning the Present Contribution

The contribution of this paper is a meta-theoretical constraint. Unlike BFO and DOLCE, it does not propose a specific taxonomy of entities; rather, it defines the informational boundary that any such taxonomy must respect to remain neutral.

Recent work has questioned the stability of top-level ontological commitments in emerging technical domains. Köhler and Neuhaus argue that large language models exhibit a *mercurial* ontological status: their classification shifts depending on task framing, deployment context, and explanatory purpose (Köhler and Neuhaus 2025). This observation is consistent with the present result by illustrating that even top-level ontological categories cannot be guaranteed to remain stable when interpretive predicates are embedded, reinforcing the need, where neutrality across uses is required, for a neutrality-preserving substrate beneath such shifting conceptualizations.

Recent work also emphasizes that ontology development itself requires negotiating consensus among stakeholders with divergent commitments (Neuhaus and Hastings 2022). This observation reinforces the present argument by contrast: while temporary disagreement during ontology construction may be instrumental in converging on a sound design, ontologies intended to operate after construction in contexts of valid and persistent interpretive disagreement among admissible frameworks cannot rely on consensus as a foundational design assumption. Instead, such ontologies must, in these contexts, be structured to accommodate disagreement structurally without requiring its resolution. The theorem specifies the formal constraints such accommodation must satisfy.

Recent ontology engineering work has emphasized tool-supported maintenance and rapid evolution in fast-moving domains. For example, the Artificial Intelligence Ontology (AIO) employs LLM-assisted curation and automated ODK pipelines to keep a domain ontology current as AI concepts and ethical concerns evolve (Joachimiak et al. 2025). Such approaches highlight the importance of distinguishing between evolving domain-level ontologies and the comparatively stable substrates on which they depend: the present result clarifies why neutrality constraints must apply to the latter even as the former necessarily change.

By grounding this constraint in the Method of Levels of Abstraction, the debate shifts from a sociological discussion of agreement to a formal requirement for observational invariance.

This result is complementary to prior work rather than competitive. Upper ontologies may inform the categorical structure of a neutral substrate. Legal and causal frameworks may operate as interpretive layers atop it. What the present work provides is a principled boundary: a specification of what must be externalized in order for the substrate to support accountability across disagreement.

3 Formal Requirements for Ontological Neutrality

This section formalizes the neutrality requirements introduced in Section 1. All formal results in subsequent sections are conditional on the definitions, assumptions, and scope conditions stated here. Unless otherwise noted, all entailment ($\vdash$) is with respect to classical first-order logic with equality.

The conceptual architecture is summarized in Figure 1.

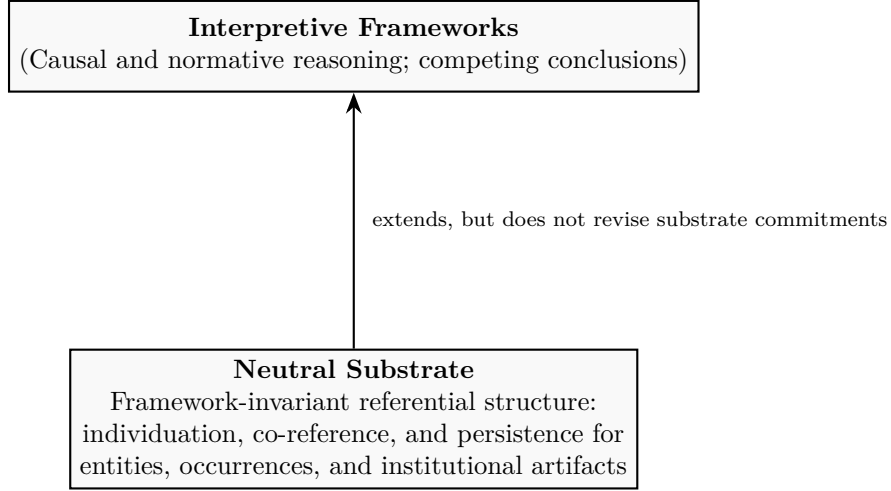


Figure 1: Conceptual architecture of a neutral substrate and interpretive frameworks.

The interface between layers consists of shared reference to entities, occurrences, and institutional artifacts, together with the referential regimes by which they are individuated, co-referred, and tracked across time. Interpretive frameworks extend this structure without revising substrate-layer commitments.

The notion of neutrality used in this paper is deliberately narrow and operational.

It does not refer to moral neutrality, political impartiality, epistemic skepticism, or the absence of content. Rather, neutrality is defined relative to the functional role an ontology plays when it is used as a shared substrate across divergent interpretive frameworks.

An ontological substrate is neutral, in the sense relevant here, if it satisfies two requirements: interpretive non-commitment and extension stability. These requirements are independent but jointly necessary. Together, they characterize the minimal conditions under which a shared ontology can support accountability without foreclosing disagreement.

Throughout this section, all requirements are understood as analytic consequences of this role-relative definition of neutrality, not as general design prescriptions for ontologies.

Interpretive non-commitment constrains what the substrate asserts at its chosen Level of Abstraction. Extension stability is a global consistency condition: even a substrate that is silent on contested predicates can fail neutrality if it encodes background axioms that conflict with some admissible interpretive framework.

3.1 Substrate-Layer Commitment

Definition 3.1 (Substrate-Layer Commitment). A substrate $\mathcal{S}$ *commits to* a proposition p when $\mathcal{S} \vdash p$; that is, when p is asserted by the substrate as holding independently of any particular interpretive framework. A *causal or normative commitment* is a substrate-layer commitment whose content is a causal relation or normative proposition. Such commitments do not include the representation of causal or normative propositions as attributed, provenance-bearing, reified assertions.

3.2 Referential Regimes

A neutral substrate is not empty. It must provide stable reference to the entities, occurrences, and institutional artifacts about which interpretive frameworks reason. This requires the substrate to fix a referential structure that remains invariant across the admissible frameworks under consideration.

Definition 3.2 (Referential Regime). A *referential regime* is the set of conditions under which a substrate individuates an entity or occurrence, determines when

two references co-refer to the same entity or occurrence, and tracks that entity or occurrence across time, records, jurisdictions, or interpretive contexts. A referential regime therefore comprises three components: *individuation*, *co-reference*, and *persistence*.

Referential regimes provide the substrate-level structure over which interpretive frameworks operate. They specify what counts as one entity or occurrence, when different descriptions or records refer to the same one, and under what conditions that referent is treated as persisting across time or context.

This terminology separates the referential role of the substrate from the interpretive role of higher layers. The substrate may fix a referential regime for an occurrence without asserting why the occurrence happened, whether it caused another occurrence, whether an actor was responsible for it, or whether it constituted a violation. Those latter claims belong to interpretive frameworks.

The neutrality theorem therefore presupposes *referential invariance*: the relevant referential regimes must be fixable at the substrate layer and must not themselves be contested across the admissible frameworks under consideration. Where individuation, co-reference, or persistence is itself contested, neutrality in the sense defined here is unavailable until that contestation is resolved or scoped out.

3.3 Admissible Frameworks

Admissibility is determined by the domain of application, not by the substrate. A framework is admissible if recognized stakeholders in the domain accept it as a legitimate interpretive stance, for example, a jurisdiction’s legal code, a scientific community’s causal methodology, or an institution’s normative charter. The substrate must accommodate all such frameworks; it does not adjudicate among them.

Definition 3.3 (Admissible Framework). A framework $\mathcal{F}$ is *admissible* if it is internally consistent ($\mathcal{F} \not\models \perp$) and represents a legitimate interpretive stance within the domain of application. The set of all admissible frameworks is denoted $\mathbb{F}$. Admissible frameworks need not be mutually compatible.

The theory treats admissibility as a domain-governance parameter, not as a property determined internally by the substrate ontology. The substrate is not a court,

regulator, or scientific authority. It does not decide which framework is correct. It provides a shared referential base over which admissible frameworks may reason, including when those frameworks reach incompatible conclusions.

3.4 Framework-Variant Propositions

The first neutrality requirement concerns propositions whose truth depends on the interpretive framework.

Definition 3.4 (Framework-Variant Proposition). Let $\mathcal{S}$ denote the substrate ontology whose neutrality properties are under analysis. A proposition p is *framework-variant* if there exist admissible frameworks $\mathcal{F}_1, \mathcal{F}_2 \in \mathbb{F}$ such that $\mathcal{S} \cup \mathcal{F}_1 \vdash p$ and $\mathcal{S} \cup \mathcal{F}_2 \vdash \neg p$.

Framework-variant propositions include, in the domains considered here, causal and normative conclusions whose truth conditions depend on legal standards, causal models, institutional rules, or other interpretive commitments. Examples include claims that one event caused another, that an actor was responsible for an occurrence, or that an action was permitted, prohibited, obligatory, compliant, noncompliant, justified, or violative.

The governing distinction is between asserting a proposition as a substrate-layer fact and representing that a framework has made the proposition as a claim. The exclusion is at the level of assertion, not vocabulary. Causal and normative predicates may appear as the opaque content of reified assertions, reports, or provenance-bearing claim objects. What neutrality excludes is asserting those predicates of substrate individuals as substrate-layer facts.

Assumption 3.5 (Contestability of Causal and Normative Propositions). *In the domains to which the neutrality requirements are applied, causal and normative propositions are contestable across admissible frameworks. In particular, if a substrate $\mathcal{S}$ asserts a causal or normative proposition p as a substrate-layer commitment, then there exists an admissible framework $\mathcal{F} \in \mathbb{F}$ such that $\mathcal{S} \cup \mathcal{F} \vdash \neg p$.*

This assumption is a scope condition on the theorem. It does not claim that causation is unreal, that normative judgment is arbitrary, or that every domain contains an active dispute about every claim. It states that, in the class of domains addressed by this paper, causal and normative conclusions are not guaranteed to

remain invariant across admissible interpretive frameworks. Accordingly, asserting such conclusions at the substrate layer risks converting the substrate into one framework’s interpretation.

3.5 Interpretive Non-Commitment

The first neutrality requirement is interpretive non-commitment. A substrate ontology satisfies this requirement if it does not assert any proposition whose truth value can vary across admissible interpretive frameworks.

Definition 3.6 (Interpretive Non-Commitment). A substrate ontology $\mathcal{S}$ satisfies *interpretive non-commitment* if it does not assert any framework-variant proposition: there is no proposition p such that $\mathcal{S} \vdash p$ and p is framework-variant with respect to $\mathbb{F}$.

Accountability systems routinely involve disagreements about what occurred, why it occurred, and how it should be evaluated. Legal frameworks may disagree about whether an action was permitted or prohibited; analytic frameworks may disagree about whether one event caused another; political frameworks may disagree about responsibility or intent. These disagreements are not merely epistemic gaps to be filled with additional data. They reflect differences in background assumptions, governing rules, and interpretive commitments.

A neutral substrate, under this definition, must therefore refrain from settling such questions at the substrate layer. If a substrate ontology asserts that a particular action was obligatory, forbidden, permitted, compliant, or violative, it commits the ontology to a particular normative framework. Similarly, if it asserts that one event caused another, it commits the ontology to a particular causal model or explanatory framework. This does not imply that causation is subjective or unreal. Rather, causal conclusions depend on explanatory frameworks such as statistical, mechanistic, or legal standards of causation that may legitimately differ across admissible frameworks. In both cases, asserting such conclusions as substrate-layer commitments violates interpretive non-commitment.

Interpretive non-commitment does not require the substrate ontology to ignore normative or causal discourse. Rather, it requires that the substrate’s own assertions remain restricted to framework-invariant referential structure. Causal and normative

claims may be represented as claims, with attribution and provenance, but their truth is evaluated only within interpretive frameworks layered above the substrate.

3.6 Extension Stability

The second neutrality requirement is extension stability. A substrate ontology satisfies this requirement if it remains consistent when extended separately by each admissible interpretive framework.

Definition 3.7 (Extension Stability). A substrate ontology $\mathcal{S}$ satisfies *extension stability* if for all admissible frameworks $\mathcal{F} \in \mathbb{F}$, $\mathcal{S} \cup \mathcal{F} \not\vdash \perp$.

Extension stability follows from the functional role of a neutral substrate, which must remain usable by independently developed interpretive frameworks whose conclusions may diverge and whose development cannot be centrally coordinated.

Extension stability does not require that all admissible interpretive frameworks be simultaneously combined into a single globally consistent theory. Rather, the requirement is *pairwise compatibility with the substrate*: for each admissible framework $\mathcal{F} \in \mathbb{F}$ considered independently, the combined theory $\mathcal{S} \cup \mathcal{F}$ must remain consistent. The substrate need not reconcile frameworks with one another. It must avoid assertions that require revision when an admissible framework disagrees.

The substrate is permitted to be extended by additional structures, records, or interpretive layers, provided such extensions do not retract, revise, or contradict substrate-layer assertions. A violation occurs when accommodating an admissible framework requires modification of the substrate itself. Disagreement between frameworks, including disagreement about causal, normative, or explanatory conclusions, is expected and permitted. What is prohibited is embedding substrate-layer commitments that make such disagreement impossible to accommodate without revising the substrate.

Consider a substrate ontology $\mathcal{S}$ used by two interpretive frameworks, $\mathcal{F}_1$ and $\mathcal{F}_2$, that contradict each other ($\mathcal{F}_1 \cup \mathcal{F}_2 \vdash \perp$). For $\mathcal{S}$ to remain stable, it must be separately consistent with each framework: $\mathcal{S} \cup \mathcal{F}_1 \not\vdash \perp$ and $\mathcal{S} \cup \mathcal{F}_2 \not\vdash \perp$. The substrate need not reconcile the frameworks. It must simply avoid assertions that either framework rejects.

Now suppose $\mathcal{S}$ commits to a causal or normative proposition p , and some admissible

framework $\mathcal{F}_2$ rejects it ($\mathcal{S} \cup \mathcal{F}_2 \vdash \neg p$). Since $\mathcal{S} \vdash p$, the combined theory entails both p and $\neg p$: $\mathcal{S} \cup \mathcal{F}_2 \vdash \perp$. The substrate must then be revised to accommodate $\mathcal{F}_2$, violating extension stability.

Now suppose $\mathcal{S}$ commits to a causal or normative proposition p . Such a substrate cannot function as a neutral foundation for the class of frameworks that includes such rejection, because accommodating the rejecting framework would require revision of the substrate.

3.7 Levels of Abstraction and Organization

The neutrality requirements constrain both the Level of Abstraction (LoA) and the Level of Organization (LoO) of the architecture.

In the Method of Levels of Abstraction, an LoA is determined by the observables made available for representation (Floridi 2008 2011). For a substrate ontology, these observables correspond to the predicate symbols and assertion forms available at the foundational layer. Neutrality requires that the substrate’s own assertions be restricted to framework-invariant referential predicates: predicates supporting referential regimes, including individuation, co-reference, persistence, provenance, temporal ordering, participation, containment, and other structural relations that assert neither causal efficacy nor normative force.

This is not a prohibition on mentioning causal or normative content. Causal and normative predicates may occur within reified assertions as opaque content, for example $\text{Asserts}(\mathcal{F}, \varphi)$, where φ is a causal or normative claim. The substrate may assert that a framework made a claim. It may not assert the claim’s truth as a substrate-layer fact.

The LoO characterizes the architectural hierarchy of the system: the ordering of layers by direction of dependence. The neutral substrate occupies the foundational organizational layer. Higher interpretive layers may extend the substrate by adding causal models, normative rules, legal standards, or evaluative conclusions. They may not mutate, retract, contradict, or revise substrate-layer commitments.

This directional dependence is essential. Identity, co-reference, and persistence are established at the substrate layer before interpretive frameworks are applied. Explanation, evaluation, and judgment operate on referents whose referential regimes

have already been fixed. Changes in higher-level interpretive logic must therefore produce new interpretive assertions or revised framework-level conclusions, not revisions to the substrate itself.

3.8 Scope of Neutrality

Neutrality, as defined here, is not global. A neutral substrate is not neutral with respect to the referential regimes it asserts. It must take substantive positions on individuation, co-reference, and persistence, because accountability requires stable referents. These commitments are necessary for accountability and do not threaten neutrality insofar as they are fixed as part of the shared substrate scope and remain invariant across the admissible frameworks under consideration.

Neutrality is therefore scoped specifically to interpretive commitments: those aspects of representation whose truth depends on legal, normative, causal, or analytic frameworks that may legitimately disagree. This restriction concerns what the substrate may assert, not whether causal or normative relations exist in the world. Such relations may be represented and reasoned about within interpretive frameworks layered above the substrate. The exclusion of such commitments from the substrate is not a matter of preference or conservatism; it is a functional requirement derived from the role the ontology is intended to play as a neutral substrate across divergent interpretive frameworks.

A further scope condition follows. The neutrality requirements assume that the substrate's framework-invariant referential structure, including its referential regime, is itself invariant across admissible frameworks. If frameworks disagree about individuation, co-reference, or persistence—for example, whether two references denote the same actor, whether an occurrence should be treated as one event or many, or whether a particular institutional artifact persists across legal or organizational change—then the substrate cannot be neutral with respect to those frameworks until that referential contestation is resolved or scoped out. This is not a defect of the framework but a boundary condition: neutrality presupposes a shared ontological ground on which interpretation can vary.

Together, interpretive non-commitment and extension stability define the functional boundaries of a neutral substrate. These requirements show that neutrality is not a lack of content, but a categorical constraint on where interpretive commitments may occur. The next section establishes that, under the contestability assumption

and referential-invariance scope condition stated here, a substrate ontology satisfies neutrality if and only if its foundational layer is restricted to framework-invariant referential structure and makes no causal or normative commitments.

4 Formal Development of the Ontological Neutrality Theorem

Section 3 defined the neutrality requirements governing a substrate ontology: interpretive non-commitment and extension stability. It also introduced the scope assumptions under which those requirements apply: causal and normative propositions are contestable across admissible frameworks, while the substrate’s framework-invariant referential structure, including its referential regime, is assumed to be invariant across the admissible frameworks under consideration.

This section establishes the relationship between those requirements and the structure of the substrate layer. Specifically, it proves that the two neutrality requirements place a structural constraint on the substrate layer: under the contestability assumption and the referential-invariance scope condition, a substrate satisfies interpretive non-commitment and extension stability if and only if its foundational layer is restricted to framework-invariant referential structure and makes no causal or normative commitments.

The argument is structural and logical rather than empirical. It follows from the interaction between interpretive disagreement and substrate-layer assertion. All necessity and sufficiency claims in this section are relative to the definitions, assumptions, and scope conditions introduced in Section 3.

4.1 Neutrality as Compatibility Across Admissible Frameworks

By definition, a neutral substrate must remain compatible with multiple admissible interpretive frameworks, even when those frameworks disagree. Compatibility here is understood in the sense of extension stability: the substrate must remain logically consistent when extended by each admissible interpretive framework, without revision or retraction of substrate-layer assertions.

Admissible frameworks may disagree about causal or normative conclusions concerning the same referents while remaining internally consistent. For example, an admissible framework may correspond to a recognized legal jurisdiction, an established causal modeling tradition, or a coherent institutional or normative charter. Admissibility is not mutual compatibility. Two admissible frameworks may contradict one another, as when two jurisdictions reach opposite legal conclusions or two causal models attribute responsibility differently. The set $\mathbb{F}$ is therefore characterized by domain-governed admissibility and internal consistency, not by global agreement among its members.

This neutrality requirement immediately constrains what the substrate may assert as foundational fact. If the substrate asserts a proposition whose truth depends on a particular causal or normative interpretation, then that assertion may conflict with an admissible framework that rejects the proposition. Because neutrality requires the substrate to remain consistent when extended by every admissible framework, such an assertion violates the neutrality requirements.

This restriction does not prohibit the representation of causal or normative concepts. A neutral substrate may record causal or normative claims as reified assertions carrying attribution and provenance without asserting their content as substrate-layer facts. For example, the substrate may assert $\text{Asserts}(\mathcal{F}, \varphi)$, where φ is a causal or normative claim, without asserting φ itself. What neutrality excludes is the substrate committing to causal or normative propositions about substrate individuals.

4.2 Framework-Contestability and Substrate Incompatibility

The arguments in the following subsections share a common structure, stated here as a general lemma.

Lemma 4.1 (Framework-Contestability Lemma). *Let $\mathcal{S}$ be a substrate ontology intended to satisfy the neutrality requirements, and let p be a framework-variant proposition with respect to $\mathbb{F}$. If $\mathcal{S}$ commits to p , then $\mathcal{S}$ violates both interpretive non-commitment and extension stability.*

Proof. Since p is framework-variant with respect to $\mathbb{F}$, there exist admissible frameworks $\mathcal{F}_1, \mathcal{F}_2 \in \mathbb{F}$ such that $\mathcal{S} \cup \mathcal{F}_1 \vdash p$ and $\mathcal{S} \cup \mathcal{F}_2 \vdash \neg p$.

Since $\mathcal{S}$ commits to p , Definition 3.1 gives $\mathcal{S} \vdash p$. Thus $\mathcal{S}$ asserts a framework-variant proposition, violating interpretive non-commitment (Definition 3.6).

Moreover, because $\mathcal{S} \vdash p$, every extension of $\mathcal{S}$ preserves p . In particular, $\mathcal{S} \cup \mathcal{F}_2 \vdash p$. But by framework-variance, $\mathcal{S} \cup \mathcal{F}_2 \vdash \neg p$. Hence

$$\mathcal{S} \cup \mathcal{F}_2 \vdash \perp.$$

Thus $\mathcal{S}$ also violates extension stability (Definition 3.7). □

The lemma identifies the general failure mode: a substrate loses neutrality when it commits to a proposition that admissible frameworks may contest. The next two subsections instantiate this structure for normative and causal commitments.

4.3 Normative Commitments

Normative conclusions are framework-dependent in the sense of Lemma 4.1. Their truth conditions depend on legal regimes, institutional authorities, temporal scope, governing rules, and interpretive stance. It is therefore a routine and expected feature of accountability systems that admissible frameworks disagree about normative conclusions.

By Assumption 3.5, if $\mathcal{S}$ commits to a normative proposition n , then n is not guaranteed invariant across admissible frameworks: there exists some $\mathcal{F} \in \mathbb{F}$ such that $\mathcal{S} \cup \mathcal{F} \vdash \neg n$. Since $\mathcal{S} \vdash n$, the proposition is framework-variant in the sense relevant to the neutrality requirements. By Lemma 4.1, the substrate violates interpretive non-commitment and extension stability.

The point is not that normative judgment is illegitimate or irrelevant to accountability. Rather, normative judgment belongs to interpretive frameworks. If the substrate asserts that an action was permitted, prohibited, obligatory, compliant, noncompliant, justified, or violative, it embeds a framework-dependent conclusion as a substrate-layer fact. Any admissible framework that rejects that conclusion cannot be layered atop the substrate without contradiction or substrate revision. Such a substrate no longer functions as a neutral substrate for the relevant framework space.

4.4 Causal Commitments

Causal attributions are likewise framework-dependent in the sense relevant to Lemma 4.1. Claims such as $\text{Caused}(e_1, e_2)$ depend on background assumptions about causal mechanisms, variable selection, counterfactual reasoning, temporal scope, intervention conditions, and model boundaries. Distinct causal frameworks may be admissible while disagreeing about specific causal relationships.

By Assumption 3.5, if $\mathcal{S}$ commits to a causal proposition c , then c is not guaranteed invariant across admissible frameworks: there exists some $\mathcal{F} \in \mathbb{F}$ such that $\mathcal{S} \cup \mathcal{F} \vdash \neg c$. Since $\mathcal{S} \vdash c$, the proposition is framework-variant in the sense relevant to the neutrality requirements. By Lemma 4.1, the substrate violates interpretive non-commitment and extension stability.

This argument does not rely on the presence of an active dispute over a specific event. It concerns the status of causal attribution as a substrate-layer assertion. As soon as a substrate asserts $\text{Caused}(e_1, e_2)$ as a foundational fact, it commits to a particular causal interpretation of the relation between e_1 and e_2 . An admissible framework that treats the two occurrences as merely correlated, as effects of a common cause, or as causally related only under different model assumptions cannot be layered atop the substrate without conflict.

To be clear, *pre-causal does not mean acausal*. The substrate does not deny that causation exists nor does it deny that causal reasoning is necessary. Rather, it refrains from embedding causal conclusions as foundational commitments. Causation is externalized to interpretive layers where causal propositions are explicit, contestable, and attributable. Causation is externalized, not eliminated.

4.5 Example: Incident Investigation

The following example illustrates the distinction between substrate-layer referential structure, framework-level conclusions, and reified assertions about claims.

Example 4.2 (Interpretive Disagreement in Incident Investigation). Consider a substrate $\mathcal{S}$ representing a workplace incident. The substrate asserts only framework-invariant referential structure, including a referential regime for the relevant agents and occurrences.

For example, the substrate may assert:

- $\text{Agent}(a)$: a is an agent;
- $\text{Occurrence}(e_1)$ and $\text{Occurrence}(e_2)$: e_1 and e_2 are occurrences;
- $\text{Participant}(a, e_1)$: a participated in e_1 ;
- $\text{Before}(e_1, e_2)$: occurrence e_1 preceded occurrence e_2 ;
- referential-regime assertions fixing the individuation, co-reference, and persistence conditions for a , e_1 , and e_2 .

The substrate does not assert $\text{Caused}(e_1, e_2)$, $\text{Responsible}(a, e_2)$, or any normative conclusion about whether a violated, satisfied, or breached an obligation.

Two admissible interpretive frameworks extend $\mathcal{S}$:

$\mathcal{F}_1$, an engineering analysis, applies a mechanistic causal model and derives $\text{Caused}(e_1, e_2)$ and $\text{Responsible}(a, e_2)$.

$\mathcal{F}_2$, a legal framework, applies jurisdiction-specific standards of responsibility and derives $\neg\text{Responsible}(a, e_2)$.

The two frameworks are mutually inconsistent with respect to the responsibility conclusion: $\mathcal{F}_1 \cup \mathcal{F}_2 \vdash \perp$. However, each is separately consistent with the substrate: $\mathcal{S} \cup \mathcal{F}_1 \not\vdash \perp$ and $\mathcal{S} \cup \mathcal{F}_2 \not\vdash \perp$. The substrate remains neutral because it provides a stable referential base without asserting either framework's conclusion as a substrate-layer fact.

Violation case. Suppose instead that $\mathcal{S}$ asserts $\text{Responsible}(a, e_2)$ as a substrate-layer fact. Then $\mathcal{S} \cup \mathcal{F}_2$ entails both $\text{Responsible}(a, e_2)$ and $\neg\text{Responsible}(a, e_2)$. Thus $\mathcal{S} \cup \mathcal{F}_2 \vdash \perp$, and the substrate violates extension stability. The failure is not that responsibility is irrelevant. The failure is that the substrate asserted a framework-level conclusion as a substrate-layer fact.

Similarly, if $\mathcal{S}$ asserts $\text{Caused}(e_1, e_2)$ as a substrate-layer fact, then any admissible framework that rejects that causal attribution cannot extend the substrate without contradiction. The substrate has embedded a causal interpretation rather than merely providing the referential structure over which causal interpretations may be made.

Reification case. Now suppose $\mathcal{S}$ neither asserts nor denies $\text{Caused}(e_1, e_2)$ or $\text{Responsible}(a, e_2)$, but instead represents the engineering framework's conclusions as reified assertions:

$$\text{Asserts}(\mathcal{F}_1, \text{Caused}(e_1, e_2))$$

and

$$\text{Asserts}(\mathcal{F}_1, \text{Responsible}(a, e_2))$$

These are substrate-layer facts about the existence, source, and attribution of claims. They are not substrate-layer commitments to the truth of the claims' contents. Both $\mathcal{F}_1$ and $\mathcal{F}_2$ can be layered atop $\mathcal{S}$ without contradiction: each framework reasons over the same referents while deriving its own conclusions.

This is the structurally neutral pattern. The substrate records discourse about the world without asserting contested conclusions about the world.

4.6 Reification and Contextual Modeling

The reification case in Example 4.2 illustrates a broader point about contextual modeling approaches. Representing causal or normative claims as reified assertions attributed to agents, institutions, documents, or frameworks, is consistent with neutrality, provided the substrate does not also assert the reified content as a substrate-layer fact.

This pattern is compatible with DOLCE/DUL-style modeling of situations and descriptions (Borgo et al. 2022a). In such approaches, a description or situation may be represented as an entity in the ontology, with attribution, provenance, temporal scope, and other metadata. The substrate may assert that a description exists, that it was made by a particular source, or that it concerns particular referents. It need not assert that the description is true.

Reification preserves interpretive non-commitment only if the substrate refrains from simultaneously asserting the reified content as a substrate-layer truth. If a causal or normative relation is asserted as a substrate-layer fact and also represented as a reified claim, neutrality is already lost. The reified representation does not cure the substrate-layer assertion. It merely duplicates it.

Thus, reification and contextual modeling do not circumvent the theorem's con-

straint. They satisfy it when they externalize interpretive commitments rather than embedding them under a different representational form. The constraint concerns what is asserted as foundational fact, not the representational mechanism through which interpretive discourse is recorded.

4.7 The Ontological Neutrality Theorem

The preceding observations yield the central result of this work.

Theorem 4.3 (Ontological Neutrality Theorem). *Let $\mathcal{S}$ be a substrate ontology intended to function as a neutral substrate across admissible interpretive frameworks, and let $\mathbb{F}$ be the set of all admissible frameworks. Suppose the domain satisfies the contestability assumption for causal and normative propositions and the referential-invariance scope condition that the substrate’s framework-invariant referential structure, including its referential regime, is not contested across the admissible frameworks under consideration. Then $\mathcal{S}$ satisfies the neutrality requirements of interpretive non-commitment and extension stability if and only if its foundational layer is restricted to framework-invariant referential structure and does not assert causal or normative conclusions as substrate-layer facts.*

Here a *causal or normative commitment* means the assertion of a causal relation or normative proposition as a substrate-layer fact, for example that one event caused another, that an actor was responsible for an occurrence, or that an action was permitted, prohibited, obligatory, compliant, noncompliant, justified, or violative. It does not include the representation of such propositions as attributed, provenance-bearing, reified assertions.

Proof. Necessity. Suppose $\mathcal{S}$ satisfies the neutrality requirements. Assume, for contradiction, that $\mathcal{S}$ commits to a causal or normative proposition p . so $\mathcal{S} \vdash p$. By Assumption 3.5, there exists an admissible framework $\mathcal{F} \in \mathbb{F}$ such that $\mathcal{S} \cup \mathcal{F} \vdash \neg p$. Since $\mathcal{S} \vdash p$, it follows that $\mathcal{S} \cup \mathcal{F} \vdash p$. Therefore $\mathcal{S} \cup \mathcal{F} \vdash p \wedge \neg p$, so $\mathcal{S} \cup \mathcal{F} \vdash \perp$. This contradicts extension stability. Moreover, because p is rejected by an admissible framework while asserted by the substrate, p is framework-variant in the relevant sense, so asserting it also violates interpretive non-commitment. Therefore, no neutral substrate may make causal or normative commitments.

It remains to identify what may remain in the foundational layer. By the referential-invariance scope condition, the substrate may assert framework-invariant referential

structure: the referential regimes by which entities, occurrences, and institutional artifacts are individuated, co-referred, and tracked across time, together with structural relations that assert neither causal efficacy nor normative force. Thus, under the stated scope condition, a neutral substrate’s foundational layer must be restricted to framework-invariant referential structure and must exclude causal and normative commitments.

Sufficiency. Suppose $\mathcal{S}$ is restricted at the foundational layer to framework-invariant referential structure and makes no causal or normative commitments. Let $\mathcal{F} \in \mathbb{F}$ be any admissible framework.

Because the assertions of $\mathcal{S}$ are, by hypothesis, framework-invariant referential assertions, and because the framework-invariant referential structure $\mathcal{S}$ asserts, including its referential regime, is, by the scope condition, not contested across admissible frameworks, no assertion of $\mathcal{S}$ is contradicted by $\mathcal{F}$. Therefore $\mathcal{S} \cup \mathcal{F} \not\models \perp$. Since $\mathcal{F}$ was arbitrary, $\mathcal{S}$ satisfies extension stability.

It remains to show interpretive non-commitment. Since $\mathcal{S}$ asserts only framework-invariant referential structure, there is no proposition p such that $\mathcal{S} \vdash p$ and p is framework-variant with respect to $\mathbb{F}$. Causal and normative propositions may appear only as reified, attributed, provenance-bearing content. The substrate may assert that a framework made a claim; it does not assert the claim’s truth. Thus $\mathcal{S}$ satisfies interpretive non-commitment.

Therefore $\mathcal{S}$ satisfies both neutrality requirements: interpretive non-commitment and extension stability.

Scope note. The sufficiency direction depends on the referential-invariance scope condition. If admissible frameworks disagree about individuation, co-reference, or persistence—for example, whether two records denote the same entity, whether an occurrence is one event or many, or whether an institutional artifact persists across legal change—then the substrate cannot be neutral with respect to those frameworks until the referential contestation is resolved or scoped out. The theorem applies to domains in which the substrate’s framework-invariant referential structure, including its referential regime, can be fixed prior to interpretive extension. It does not claim that all domains admit neutral substrates. $\square$

This result is not contingent on a particular ontology, syntax, protocol, or imple-

mentation technology. It follows directly from the functional role the substrate is required to play under persistent interpretive disagreement. Neutrality across disagreement is structurally incompatible with substrate-layer causal or normative commitments.

The next section considers the implications of this result for ontology design, clarifying the permitted and excluded substrate-layer commitments and the mechanisms by which interpretation, evaluation, and explanation may be externalized without loss of accountability.

5 Implications for Ontology Design

The result established in Section 4 does not prescribe a particular ontology. Rather, it constrains the design space of any ontology intended to function as a neutral substrate across admissible interpretive frameworks under persistent disagreement. This section clarifies what such constraints require, what they exclude, and how interpretation, evaluation, and explanation may be accommodated without violating neutrality.

The governing distinction is the one established in the previous sections: a neutral substrate may represent that a claim was made, by whom, and with what provenance, but it may not assert the truth of causal or normative claims as substrate-layer facts. The exclusion is therefore at the level of assertion, not vocabulary. Causal and normative content may appear as reified, attributed, provenance-bearing discourse; it may not be asserted of substrate individuals as a foundational ontological commitment.

5.1 Substrate Exclusions

The primary implication is negative but precise. A neutral ontological substrate must exclude, at the foundational layer, assertions that are framework-variant across admissible frameworks. In particular, the substrate must not assert causal or normative conclusions as substrate-layer facts.

Normative exclusions include substrate-layer assertions that an action, event, actor, institution, or artifact was permitted, prohibited, obligatory, compliant, noncompli-

ant, justified, violative, blameworthy, liable, or responsible in a framework-dependent sense. Such conclusions presuppose legal, political, institutional, or evaluative standards that may vary across admissible frameworks.

Causal exclusions include substrate-layer assertions that one event caused, produced, prevented, enabled, was responsible for, or explains another. Such conclusions presuppose causal models, counterfactual assumptions, variable selections, mechanistic accounts, evidentiary thresholds, or legal standards of causation that may vary across admissible frameworks.

These exclusions do not deny the importance of causal or normative reasoning for accountability. Rather, they preserve the distinction between representation and interpretation. Embedding causal or normative conclusions as substrate-layer facts collapses that distinction, commits the substrate to a particular interpretive framework, and thereby violates neutrality under the theorem.

The exclusions apply specifically to asserted conclusions, not to the referents about which such conclusions are made. Actors, occurrences, institutional artifacts, jurisdictions, records, documents, claims, reports, and provenance metadata may all be represented without committing to how they are causally explained or normatively evaluated. The substrate may therefore be rich in referential structure while remaining silent on contested interpretation.

5.2 Substrate Requirements

Although the constraints are restrictive, they do not leave the ontology empty or impoverished. On the contrary, a neutral substrate must make strong, explicit commitments in areas that are invariant across the admissible frameworks under consideration. These commitments provide the shared referential ground on which interpretive disagreement can occur.

In terms of the Method of Levels of Abstraction (LoA) (Floridi 2008 2011), the substrate fixes a framework-invariant referential structure: the predicate symbols and assertion forms needed to establish referential regimes for entities, occurrences, and institutional artifacts. A referential regime comprises the substrate-level conditions of individuation, co-reference, and persistence. It determines what counts as one referent, when two references denote the same referent, and how that referent is tracked across time, records, jurisdictions, or interpretive contexts.

The claim that a neutral substrate relies on such structure does not assert that all questions of individuation, co-reference, persistence, or boundary demarcation are framework-independent. Many such questions may be legitimately contested: the spatial or temporal granularity of an occurrence, the criteria for organizational continuity, the recognition of informal institutions, or the question whether two records refer to the same actor. Where such referential questions are contested across admissible frameworks, the domain does not yet admit a neutral substrate with respect to those frameworks. The referential contestation must first be resolved or scoped out.

Neutrality instead requires that the substrate fix a minimal and domain-appropriate referential regime sufficient to support accountability relations. These substrate-layer commitments are invariant not because they resolve all ontological disagreement, but because they remain stable under extension by mutually incompatible causal, legal, analytic, or normative frameworks.

In particular, the substrate must provide:

- stable reference to entities, occurrences, and institutional artifacts that participate in accountability relationships;
- explicit referential regimes specifying individuation, co-reference, and persistence conditions for those referents;
- provenance and attribution structures sufficient to record claims, reports, sources, documents, and interpretive contexts;
- structural relations such as temporal ordering, participation, containment, jurisdictional association, document linkage, and record membership, insofar as these assert neither causal efficacy nor normative force.

These commitments are not optional. Accountability depends on the ability to refer stably to who acted, what occurred, which institutional instruments existed, which records were made, and within what jurisdictional or organizational scope. When fixed as part of the shared substrate scope, such commitments do not vary with interpretive stance in the way causal or normative conclusions do, and therefore do not threaten neutrality within that scoped domain.

5.3 Externalizing Interpretation Without Loss

A common concern is that excluding causal and normative commitments from the substrate renders the ontology incapable of supporting explanation, evaluation, judgment, or accountability. The result shows that the opposite is true: such functions must be externalized in order to be supported robustly across admissible disagreement.

Interpretation is accommodated by organizing the system into distinct Levels of Organization (LoO) (Floridi 2011). The foundational LoO, the substrate, records the referential ground: entities, occurrences, institutional artifacts, referential regimes, records, claims, attribution, and provenance. Higher-order LoOs record the interpretive layer: causal explanations, legal classifications, normative evaluations, responsibility assignments, compliance judgments, or analytic conclusions.

By maintaining this structural separation, multiple, mutually inconsistent interpretive layers can operate over the same foundational substrate without introducing inconsistency at the substrate layer. Disagreement, revision, and contestation are represented by the coexistence of multiple attributed interpretive assertions, not by revision of substrate-layer facts.

Formally, the relation between the foundational layer (LoO_0) and an interpretive layer (LoO_n , where $n > 0$) is a vertical stratification governed by a read-only interface constraint. An interpretive framework $\mathcal{F}$ layered at LoO_n may expand the signature by defining new causal, normative, legal, evaluative, or analytic predicates over the referents established by LoO_0 . However, this expansion constitutes a conservative extension of the substrate: the operational primitives of LoO_n are strictly prohibited from mutating, retracting, or contradicting the foundational constants, axioms, or identity mappings asserted at LoO_0 .

This directional dependence guarantees that changes or conflicts within higher-level interpretive logic do not propagate downward to destabilize the substrate. The substrate remains a stable point of reference; the interpretive frameworks supply the competing explanations and evaluations.

This architecture does not diminish the importance of causal reasoning or normative judgment. It relocates them to interpretive layers, where the relevant assumptions are explicit, attributed, contestable, and revisable. The substrate does not certify compliance, assign responsibility, or determine causation. It makes such determina-

tions adjudicable by preserving the referents, claims, sources, and provenance over which admissible frameworks may reason.

5.4 Permitted Structural Relations and Excluded Commitments

The exclusion of causal and normative commitments from the substrate layer does not entail the exclusion of all structured relations. A neutral substrate permits relations that establish referential, structural, temporal, documentary, institutional, or provenance linkage without asserting causal efficacy or evaluative force.

Examples of permitted substrate-layer relations may include:

- temporal ordering, such as one occurrence preceding another;
- participation, such as an actor participating in an occurrence;
- containment or membership, such as a record belonging to a collection;
- provenance, such as a claim being recorded by a source;
- attribution, such as a framework, agent, or document making a claim;
- jurisdictional or institutional association, insofar as the relation identifies context without asserting legal consequence;
- co-reference relations, insofar as they belong to the fixed referential regime of the substrate.

By contrast, relations that embed causal attribution or normative judgment are excluded from the substrate as substrate-layer facts. Examples include:

- $\text{Caused}(e_1, e_2)$ as a substrate assertion;
- $\text{Responsible}(a, e)$ as a substrate assertion;
- $\text{Violated}(a, r)$ as a substrate assertion;
- $\text{Compliant}(s, r)$ as a substrate assertion;
- $\text{Obligated}(a, \alpha)$ as a substrate assertion.

Such predicates may still appear as the opaque content of reified assertions, reports, or framework-level conclusions. For example, the substrate may assert: [$\text{Asserts}(\mathcal{F}, \text{Caused}(e_1, e_2))$.] This asserts that framework $\mathcal{F}$ made the causal claim. It does not assert $\text{Caused}(e_1, e_2)$ as a substrate-layer fact.

The neutrality boundary is therefore drawn not at the presence of structure, nor at the syntactic occurrence of causal or normative vocabulary, but at the embedding of framework-dependent conclusions as foundational ontological facts.

5.5 Reification as a Boundary Mechanism

Reification plays a critical role in enforcing the boundary between substrate and interpretation. By treating claims, reports, descriptions, and assertions as entities in their own right, the ontology can represent discourse about the world without asserting conclusions about the world itself.

However, reification functions as a boundary mechanism only when applied consistently. If causal or normative relations are asserted as substrate-layer facts and merely duplicated as reified claims, neutrality is not preserved. The substrate has already endorsed the content. Reification supports neutrality only when the substrate refrains entirely from asserting the reified content as true.

Properly implemented, reification transforms a framework-dependent proposition, for example $\text{Caused}(e_1, e_2)$, into an assertion individual, for example A_{72} . The substrate may then assert framework-invariant metadata about A_{72} :

$$\text{Assertion}(A_{72}), \quad \text{RecordedBy}(A_{72}, \mathcal{F}_1), \quad \text{HasContent}(A_{72}, \ulcorner \text{Caused}(e_1, e_2) \urcorner).$$

The quotation notation indicates that the causal proposition is represented as content, not asserted as a substrate-layer fact.

This type shift changes the representational status of the claim. Instead of evaluating $\text{Caused}(e_1, e_2)$ directly against the world at the substrate layer, the substrate evaluates only framework-invariant metadata about an assertion object: who made it, when it was recorded, what it concerns, what provenance attaches to it, and under which interpretive context it was made. The truth conditions of the claim remain inside the interpretive layer.

This mechanism preserves category discipline. It distinguishes:

- substrate-layer facts about assertion objects;
- framework-level conclusions contained within those assertions;
- provenance or attribution metadata about the source of the claim;

- later evaluations of the claim by admissible frameworks.

Thus, reification and contextual modeling do not evade the theorem’s constraint. They satisfy it when they externalize interpretive commitments rather than embedding them under a different description. Neutrality is not achieved by representational cleverness, but by disciplined exclusion of causal and normative commitments from substrate-layer assertion.

5.6 Accountability Preservation

Externalizing causal and normative interpretation does not weaken accountability. It preserves accountability under disagreement by allowing competing explanations and evaluations to coexist over a stable shared referential base.

In accountability contexts, different parties may reasonably disagree about what caused an outcome, whether a rule was violated, whether an actor was responsible, or whether an institutional response was justified. If the substrate embeds one of these conclusions as fact, then the record is no longer neutral with respect to the parties who reject that conclusion. The substrate becomes a site of adjudication rather than a shared basis for adjudication.

A neutral substrate instead records the referents, claims, sources, and provenance needed for later evaluation. It allows one framework to assert causation, another to deny responsibility, another to classify conduct as compliant, and another to challenge that classification, all while preserving stable reference to the same entities, occurrences, records, and institutional artifacts.

The substrate therefore does not function as a truth oracle. It does not decide whether a recorded claim is correct. It guarantees something narrower and more durable: that claims, sources, provenance, and referents are faithfully recorded and stably available for evaluation by admissible frameworks. Truth-evaluation is externalized, not abandoned.

This is the practical significance of the theorem. A substrate that refuses to adjudicate causal or normative truth at the foundational layer is not truth-indifferent. It is designed so that truth, responsibility, compliance, and causation can be evaluated without the record itself having prejudged the matter.

5.7 Summary of Structural Constraints

The structural constraints established by the theorem can be summarized succinctly. Any ontology satisfying the neutrality requirements, under the contestability assumption and the referential-invariance scope condition, must:

- restrict the foundational LoA to assertions of framework-invariant referential structure, including the individuation, co-reference, and persistence conditions by which entities, occurrences, and institutional artifacts are fixed and tracked;
- assert no framework-variant proposition at the substrate layer, including no proposition whose truth conditions depend on causal or normative interpretation, regardless of current consensus among admissible frameworks;
- preserve the LoO hierarchy so that interpretation, evaluation, and explanation are supported exclusively through externalized higher-level organizational layers that extend but do not revise the substrate;
- satisfy extension stability by remaining consistent under extension by mutually incompatible interpretive frameworks, without revision of substrate-layer commitments.

These constraints do not define a particular ontology. They delimit the structural form any ontology must take if it is to function as a neutral substrate for accountability under persistent interpretive disagreement.

The following section concludes by situating these constraints as boundary conditions on ontologies intended to function as neutral substrates across domains and applications in which the neutrality assumptions hold.

6 Conclusion

This paper has examined the structural conditions under which an ontology can function as a neutral substrate across admissible interpretive frameworks under persistent disagreement. The central problem is not semantic heterogeneity alone, nor the absence of a shared vocabulary. It is the need to maintain stable shared reference while allowing legal, causal, analytic, political, or normative frameworks to reach incompatible conclusions about the same referents.

The result established here is conditional but strong. Under the neutrality re-

quirements of interpretive non-commitment and extension stability, under the contestability assumption for causal and normative propositions, and under the scope condition that the substrate’s framework-invariant referential structure, including its referential regime, is invariant across the admissible frameworks under consideration, a substrate ontology satisfies neutrality if and only if its foundational layer is restricted to framework-invariant referential structure and does not assert causal or normative conclusions as substrate-layer facts.

This theorem identifies a categorical boundary in ontology design. A neutral substrate is not empty, contentless, or epistemically indifferent. It must assert enough structure to support accountability: entities, occurrences, institutional artifacts, referential regimes, provenance, attribution, and other framework-invariant structural relations. In particular, it must provide the referential regimes by which relevant referents are individuated, co-referred, and tracked across time, records, jurisdictions, or interpretive contexts.

At the same time, the substrate must not assert causal or normative conclusions as foundational facts. Claims that one event caused another, that an actor was responsible, that an action was permitted or prohibited, or that a system was compliant or noncompliant belong to interpretive frameworks. They may be represented by the substrate as reified, attributed, provenance-bearing claims, but their truth must not be asserted at the substrate layer. The exclusion is therefore at the level of assertion, not vocabulary.

This distinction is the core design constraint. A substrate may record that a framework made a causal or normative claim; it may record who made the claim, when it was made, what referents it concerns, and what provenance supports it. But the substrate may not convert that claim into a substrate-layer ontological fact without forfeiting neutrality. Representing discourse is compatible with neutrality. Endorsing the contested content of that discourse at the foundational layer is not. The theorem also clarifies the scope of neutrality. Neutrality is not global. A substrate is not neutral with respect to the referential regimes it asserts. It must take substantive positions on individuation, co-reference, and persistence where those positions are required for stable accountability. If admissible frameworks disagree about those referential conditions, then neutrality in the sense defined here is unavailable until the referential contestation is resolved or scoped out. This limitation is not a defect of the theorem; it is a boundary condition on the possibility of a neutral substrate.

The resulting architecture separates two tasks that accountability systems often conflate. The substrate provides stable referential ground: what was recorded, what it concerns, who asserted it, and what provenance accompanies it. Interpretive frameworks provide explanation, evaluation, judgment, responsibility assignment, and causal or normative conclusion. This separation allows competing frameworks to reason over the same substrate without forcing the substrate itself to adjudicate among them.

The practical implication is that neutral substrates should be designed as pre-causal and pre-normative foundations. Pre-causal does not mean acausal, and pre-normative does not mean normatively irrelevant. Rather, causal and normative reasoning are externalized to framework layers where their assumptions are explicit, attributed, contestable, and revisable. This preserves accountability under disagreement by making competing interpretations representable without destabilizing the shared record.

The contribution of the paper is therefore not a new ontology, nor a protocol for resolving disagreement. It is a structural constraint on any ontology intended to function as a neutral substrate in accountability contexts. Where persistent interpretive disagreement is expected, and where the substrate must remain stable across admissible frameworks, causal and normative conclusions cannot be asserted as substrate-layer facts. They must be externalized as framework-level conclusions or represented as reified claims with provenance.

In this sense, ontological neutrality is not a lack of commitments. It is a disciplined allocation of commitments across layers. The substrate commits to stable reference. Frameworks commit to interpretation. Accountability is preserved not by eliminating disagreement, but by giving disagreement a stable shared object over which it can be expressed, evaluated, and contested.

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Statements and Declarations

Author Contributions

The author was solely responsible for the conception, analysis, and writing of this manuscript.

Declaration of Conflicting Interest

The author declares no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Appendix A. Glossary of Terms

The glossaries below are explanatory aids. They do not introduce additional definitions, assumptions, or theorem conditions. Where wording differs from the formal development, Sections 3 and 4 are authoritative.

Throughout these glossaries, the exclusion of causal and normative commitments is stated at the level of substrate-layer assertion, not vocabulary. Causal and normative claims may be represented as reified, attributed, provenance-bearing content; they may not be asserted of substrate individuals as substrate-layer facts.

Core Ontology Concepts

General ontological vocabulary used in the formal framework.

Ontology Types

Ontology

A formal representation of entities within a domain and the relationships that hold between them, expressed in a logic or schema suitable for inference.

Ontology, Applied

The use of ontological methods to structure and analyze real-world domains for practical purposes. Applied ontology is the field; an applied ontology is an ontology deployed in a specific domain context.

Ontology, Domain

An ontology that represents the entities, relations, and constraints specific to a particular subject area, such as legal institutions, biomedical processes, or financial instruments. Domain ontologies are often built by extending a foundational or upper ontology with domain-specific vocabulary. Contrast with *Ontology, Foundational*.

Ontology, Epistemic

An ontology that represents claims, beliefs, assertions, reports, or descriptions as objects of representation. In the setting of this paper, epistemic modeling allows an ontology to represent that a causal or normative claim

was made, without asserting the claim’s truth as a substrate-layer fact.
Contrast with *Ontology, Metaphysical*.

Ontology, Foundational

An ontology that provides domain-independent upper-level categories intended to be extended for specific applications. Foundational ontologies define general categories such as continuants, occurrents, roles, processes, situations, or descriptions, without committing to a particular domain application. The Basic Formal Ontology (BFO) is a realist foundational ontology grounded in scientific practice and distinguishes continuants from occurrents. The Descriptive Ontology for Linguistic and Cognitive Engineering (DOLCE) has a cognitive and linguistic orientation and supports situation-based modeling of descriptions and contexts. Also called *Ontology, Upper*. Contrast with *Ontology, Domain*.

Ontology, Metaphysical

An ontology that represents the world directly, asserting what exists and what relations hold as ontological facts. In the terminology of this paper, metaphysical modeling embeds commitments about the structure of reality at the substrate layer. Contrast with *Ontology, Epistemic*.

Ontology, Upper

See *Ontology, Foundational*.

Formal Vocabulary and Assertion

Vocabulary

The set of symbols available in an ontology for representing entities, properties, relations, and assertion forms. In a formal ontology, the vocabulary consists primarily of predicate symbols together with any constants or functions used in the representation. At a given LoA, the vocabulary helps determine which distinctions can be expressed. In this paper, the central neutrality constraint concerns what is asserted as a substrate-layer fact, not whether a causal or normative symbol can occur as the content of a reified assertion.

Predicate Symbol

A symbol in the ontology’s formal vocabulary that denotes a property or relation. The syntactic occurrence of a predicate symbol is distinct

from the assertion of a proposition using that symbol. In this paper, neutrality does not require causal or normative predicate symbols to be impossible or absent from all representation. It requires that propositions using those predicates not be asserted as substrate-layer facts when they express causal or normative commitments.

Ontological Primitive

A term or predicate that an ontology takes as undefined, using it to define or constrain other terms rather than deriving it from prior definitions. In a neutral substrate, primitives used in substrate-layer assertions must support framework-invariant referential structure. Causal and normative predicates may appear as reified content, but causal and normative commitments may not be asserted as substrate-layer facts.

Ontological Assertion

The act of embedding a commitment in an ontology as a foundational fact. An ontological assertion is stronger than merely representing that some agent or framework made a claim: it treats the asserted proposition as part of the ontology itself. See *Embedding*.

Representation Mechanisms

Representation

The encoding of information within an ontology. Representation is broader than embedding: an ontology may represent a claim via reification without asserting the claim as a foundational fact. See also *Embedding* and *Reification*.

Embedding

The act of asserting a claim as a foundational fact within an ontology, such that it becomes part of the substrate layer. Embedded content is treated as true by the ontology itself, not merely recorded as a claim made by some agent, document, institution, or framework. Contrast with *Reification*.

Reification

The representation of a claim, assertion, description, or relation as an entity rather than as a direct ontological fact. Reification allows the ontology to represent discourse about the world without asserting contested conclusions

about the world itself. For example, the substrate may represent that a framework asserted a causal or normative claim, without embedding that claim as a substrate-layer fact. See also *Ontology*, *Epistemic*. Contrast with *Embedding*.

Referential Structure

Individuation

The component of a referential regime that determines what counts as one referent. Individuation specifies the conditions under which an entity, occurrence, or institutional artifact is treated as a distinct object of reference.

Co-reference

The component of a referential regime that determines when two references denote the same referent. Co-reference concerns the relation among names, descriptions, records, identifiers, or representations that point to a common entity, occurrence, or institutional artifact.

Persistence

The component of a referential regime that determines when a referent remains the same across time, records, jurisdictions, contexts, or transformations. Persistence concerns the continued identity of the referent under change.

Referential Regime

The substrate-level structure by which relevant referents are individuated, co-referred, and tracked across time or context. A referential regime comprises three components: *individuation*, *co-reference*, and *persistence*. In this paper, referential regimes appear as the scope condition required for a neutral substrate: the relevant referential regime must be fixed sufficiently to remain invariant across the admissible frameworks under consideration.

Architectural Levels

Level of Abstraction (LoA)

A perspective defined by the observables or predicate symbols made

available for representation at a given level. In this paper, the foundational LoA constrains what the substrate may assert as a substrate-layer fact. Causal and normative predicates may occur as the content of reified assertions, but causal and normative commitments may not be asserted of substrate individuals as substrate-layer facts.

Level of Organization (LoO)

The architectural layer at which entities, relations, and assertions are modeled within a system. Levels of Organization are structural rather than merely scalar: they differ by direction of dependence (for example, foundational versus interpretive), not simply by degree, granularity, or complexity. A system may have multiple LoOs arranged hierarchically, with foundational layers supporting higher-order interpretive layers. In a neutral-substrate architecture, higher interpretive layers may extend but not revise substrate-layer commitments.

Substrate

The foundational layer of an ontology that functions as a shared representational base across interpretive frameworks. Formally, the *substrate ontology* is the ontology whose foundational layer provides entities, occurrences, institutional artifacts, provenance structures, and referential regimes. For brevity, this paper refers to that ontology simply as the *substrate*. A neutral substrate does not assert causal or normative commitments as substrate-layer facts. It may represent causal or normative claims as reified, attributed, provenance-bearing content. Also called *substrate layer* or *foundational layer*.

Neutrality and Interpretation

Terms specific to the neutrality requirements defined in this paper.

Neutrality Requirements

Neutrality

The property of a substrate ontology that satisfies both interpretive non-commitment and extension stability. Neutrality, in this paper, is not moral neutrality, political impartiality, epistemic skepticism, or absence of

content. It is a role-relative structural property: the substrate must provide stable shared reference without asserting framework-variant conclusions as substrate-layer facts.

Interpretive Non-Commitment

The requirement that a substrate not assert framework-variant propositions as substrate-layer commitments. In particular, a neutral substrate must not assert causal or normative conclusions as substrate-layer facts when their truth depends on an admissible interpretive framework. Interpretive non-commitment does not prevent the substrate from representing that such claims were made; it prevents the substrate from asserting those claims as true. Together with *Extension Stability*, this requirement defines *Neutrality*.

Extension Stability

The requirement that a substrate remain logically consistent when extended separately by each admissible interpretive framework. Extension stability is pairwise: for each admissible framework $\mathcal{F}$, the combination $\mathcal{S} \cup \mathcal{F}$ must remain consistent. It does not require all admissible frameworks to be mutually consistent with one another. Together with *Interpretive Non-Commitment*, this requirement defines *Neutrality*.

Neutral Substrate

A substrate ontology that satisfies neutrality. A neutral substrate provides stable shared reference across admissible interpretive frameworks without asserting framework-variant conclusions as substrate-layer facts. It may represent causal or normative claims as reified, attributed, provenance-bearing content, but it does not assert those claims as true at the substrate layer.

Substrate-Layer Fact

A proposition asserted by the substrate ontology as holding independently of any particular interpretive framework. Substrate-layer facts are part of the shared base over which admissible frameworks may extend, reason, disagree, or derive conclusions. In a neutral substrate, substrate-layer facts must be restricted to framework-invariant referential structure and other framework-invariant structural commitments.

Framework-Level Conclusion

A proposition derived within an interpretive framework from substrate-layer facts together with the framework's own rules, assumptions, methods,

or norms. Framework-level conclusions may vary across admissible frameworks. They are not asserted by the substrate merely because they can be derived by some framework.

Interpretive Frameworks

Interpretive Framework

A coherent set of principles, rules, methods, assumptions, or norms used to evaluate, explain, classify, or assign meaning to shared referents. Examples include legal jurisdictions, causal modeling traditions, ethical theories, regulatory schemes, and analytic methodologies. Interpretive frameworks may disagree with one another while each remains internally consistent.

Admissible Framework

An interpretive framework that is internally consistent and recognized as a legitimate stance within the domain of application. Admissibility is treated in this paper as a domain-governance parameter, not as something determined by the substrate itself. Admissible frameworks need not be mutually compatible.

Pluralism

The coexistence of multiple, potentially incompatible interpretive frameworks within a single domain or system. A neutral substrate supports pluralism by providing shared referents and framework-invariant structure without committing the substrate to the conclusions of any single framework.

Accountability System

A data system designed to support explanation, evaluation, contestation, or assignment of responsibility across multiple stakeholders or interpretive perspectives. Accountability systems are a motivating application for neutral substrates because they often must accommodate persistent disagreement about causes, obligations, interpretations, and responsibility. This paper uses accountability as an application context; it does not define a specific accountable-record architecture.

Commitment and Derivation

Commitment

A proposition asserted by an ontology as holding at the substrate layer, independently of any particular interpretive framework. Commitments are the content of ontological assertions: what the ontology treats as true, not merely what has been claimed by external agents, documents, institutions, or frameworks. In a neutral substrate, framework-variant propositions may not be asserted as substrate-layer commitments. Contrast with *Conclusion*.

Conclusion

A proposition derived by an interpretive framework from substrate-layer commitments together with the framework's own rules, assumptions, methods, or norms. Conclusions belong to interpretive layers, not to the substrate. Contrast with *Commitment*.

Externalization

The architectural practice of locating contested interpretation in interpretive layers above the substrate, rather than embedding contested conclusions as substrate-layer commitments. Externalization preserves neutrality by ensuring that framework-dependent conclusions remain associated with the frameworks that derive or assert them. Reification is one mechanism for externalization.

Substrate Revision

The modification, retraction, or replacement of substrate-layer commitments in response to the introduction of a new interpretive framework. Substrate revision violates extension stability when it is required in order to accommodate an admissible framework. Under the neutrality definition, interpretive conflict must not be resolved by revising the substrate's framework-invariant commitments.

Compatibility and Scope

Consistency ($\nvdash \perp$)

The property of a set of statements such that no contradiction can be derived from its assertions. The notation $\mathcal{S} \nvdash \perp$ indicates that a substrate ontology $\mathcal{S}$ is consistent. The notation $\mathcal{S} \vdash \perp$ indicates that $\mathcal{S}$ is

inconsistent.

Compatibility

The property of a substrate ontology and an interpretive framework such that their combination does not entail contradiction. Formally, $\mathcal{S} \cup \mathcal{F} \not\models \perp$. In this paper, compatibility is assessed pairwise between the substrate and each admissible framework.

Pairwise Compatibility

The requirement that each admissible framework be separately compatible with the substrate. Pairwise compatibility does not require admissible frameworks to be compatible with one another. This distinction allows the substrate to support mutually incompatible interpretive frameworks without collapsing their disagreement.

Boundary Condition

A precondition that must hold for a framework or result to apply. In this paper, the central boundary condition is that the substrate's referential regime is fixed at the substrate layer and remains invariant across the admissible frameworks under consideration. Domains where individuation, co-reference, or persistence are themselves contested do not admit neutral substrates in the sense defined here until that contestation is resolved or scoped out. Boundary conditions delimit scope; they are not defects but explicit constraints on applicability.

Scope Condition

A condition limiting the domain over which the neutrality theorem applies. The central scope condition in this paper is referential invariance: the relevant referential regime must be fixed at the substrate layer and remain invariant across the admissible frameworks under consideration. If the referential regime itself is contested, the theorem does not claim that a neutral substrate is available until the contestation is resolved or excluded from the domain of application.

Invariant and Variant Content

Framework-Invariant Referential Structure

The referential structure whose interpretation remains stable across the admissible frameworks under consideration. In this paper, framework-invariant referential structure includes the substrate's fixed referential

regime: the individuation, co-reference, and persistence conditions by which entities, occurrences, and institutional artifacts are identified and tracked. Such structure may be asserted at the substrate layer because its interpretation does not depend on a particular admissible framework. Contrast with *Framework-Variant Proposition*.

Framework-Variant Proposition

A proposition whose truth value may differ across admissible interpretive frameworks. A framework-variant proposition may be derived, asserted, or evaluated within an interpretive framework, but it may not be asserted as a substrate-layer commitment under the neutrality definition. Formally, framework variance is defined in Definition 3.4. Contrast with *Framework-Invariant Referential Structure*.

Shared Referent

An entity, occurrence, or institutional artifact to which multiple interpretive frameworks refer through the substrate’s referential regime. Shared referents provide common ground: they allow incompatible interpretive conclusions to be recognized as conclusions about the same thing, rather than as merely equivocal uses of different references.

Disagreement

Persistent Interpretive Disagreement

The condition in which multiple admissible interpretive frameworks reach incompatible conclusions about shared referents, and the disagreement is structural rather than merely a temporary result of missing data, translation failure, or lack of coordination. Persistent interpretive disagreement is the motivating condition for neutral substrate design.

Semantic Heterogeneity

The condition in which multiple ontologies, schemas, or vocabularies exist for the same domain, and disagreement is treated primarily as a translation, alignment, mapping, or mediation problem. Semantic heterogeneity often assumes that disagreement can be resolved through better alignment. Persistent interpretive disagreement, by contrast, may remain even when reference is shared and terminology is aligned. Contrast with *Persistent Interpretive Disagreement*.

Causal and Normative Concepts

Terms relating to the classes of commitment excluded from neutral substrates.

Causal Concepts

Causal Claim

A claim that asserts or denies a causal relation among entities, occurrences, events, conditions, or processes. Examples include claims that one event caused another, that an omission prevented an outcome, or that a condition contributed to an occurrence. In a neutral substrate, causal claims may be represented as reified, attributed, provenance-bearing content, but they may not be asserted as substrate-layer facts.

Causal Predicate

A predicate symbol used to express causal relations, causal contribution, prevention, enablement, production, explanation, or dependence. The syntactic occurrence of a causal predicate is not itself prohibited by neutrality. What neutrality prohibits is the substrate-layer assertion of causal commitments using such predicates. Causal predicates may occur as the content of reified assertions.

Causal Attribution

The assignment of a causal relation between entities, occurrences, events, or conditions by an interpretive framework. Causal attribution is typically framework-dependent because different causal models may attribute influence, responsibility, contribution, or explanation differently. In a neutral substrate, causal attributions belong to interpretive layers or to reified attributed content, not to substrate-layer facts. Contrast with *Causal Commitment*.

Causal Commitment

A substrate-layer assertion that treats a causal relation as true independently of any particular interpretive framework. Examples include asserting that one occurrence caused another, that an action causally contributed to an outcome, or that a condition explains an event. Under the neutrality theorem, the substrate may not make causal commitments when their truth is contestable across admissible frameworks.

Pre-Causal

The property of a substrate whose foundational layer does not assert causal conclusions as substrate-layer facts. Pre-causal does not mean acausal, causation-denying, or unable to represent causal discourse. It means prior to causal attribution: the substrate represents entities, occurrences, provenance, temporal order, participation, and other framework-invariant structure without asserting causal conclusions as true. Causal reasoning may occur in interpretive layers, where model assumptions are explicit and disagreement remains representable. Causation is externalized, not eliminated.

Normative and Deontic Concepts**Normative Claim**

A claim that asserts or denies an evaluative, legal, moral, institutional, policy-based, or responsibility-bearing conclusion. Examples include claims that an action was permitted, prohibited, obligatory, compliant, noncompliant, justified, violative, blameworthy, liable, or responsible. In a neutral substrate, normative claims may be represented as reified, attributed, provenance-bearing content, but they may not be asserted as substrate-layer facts.

Normative Predicate

A predicate symbol used to express evaluation, obligation, permission, prohibition, compliance, violation, justification, responsibility, liability, or related normative status. The syntactic occurrence of a normative predicate is not itself prohibited by neutrality. What neutrality prohibits is the substrate-layer assertion of normative commitments using such predicates. Normative predicates may occur as the content of reified assertions.

Normative Attribution

The assignment of normative status to an entity, occurrence, action, omission, record, or institutional artifact by an interpretive framework. Normative attribution includes assigning obligations, permissions, prohibitions, violations, compliance status, responsibility, blame, liability, or justification. In a neutral substrate, normative attributions belong to interpretive layers or to reified attributed content, not to substrate-layer

facts. Contrast with *Normative Commitment*.

Normative Commitment

A substrate-layer assertion that treats an evaluative, legal, moral, institutional, policy-based, or responsibility-bearing conclusion as true independently of any particular interpretive framework. Normative commitments may reflect ethical, legal, institutional, procedural, regulatory, or policy frameworks and may therefore vary across admissible interpretive perspectives. Under the neutrality theorem, the substrate may not make normative commitments when their truth is contestable across admissible frameworks.

Deontic Claim

A normative claim specifically concerning obligation, permission, prohibition, duty, authorization, or requirement. Deontic claims state what must, may, or must not be done within some normative framework. In a neutral substrate, deontic claims may be represented as reified, attributed, provenance-bearing content, but they may not be asserted as substrate-layer facts.

Deontic Predicate

A predicate symbol used to express obligation, permission, prohibition, duty, authorization, or requirement. Deontic predicates are normative in character. Their occurrence as reified content is compatible with neutrality. Their use in substrate-layer assertions of deontic commitments is not compatible with neutrality.

Deontic Commitment

A substrate-layer assertion that treats an obligation, permission, prohibition, duty, authorization, or requirement as holding independently of any particular interpretive framework. Deontic commitments are a subset of normative commitments: they encode what ought, may, or must not be done, rather than merely what exists, occurs, or is referenced. Under the neutrality theorem, the substrate may not make deontic commitments when their truth is contestable across admissible frameworks.

Pre-Normative

The property of a substrate whose foundational layer does not assert normative conclusions as substrate-layer facts. Pre-normative does not mean non-normative, norm-denying, value-free, or unable to represent

normative discourse. It means prior to normative attribution: the substrate represents entities, occurrences, provenance, temporal order, participation, institutional artifacts, and other framework-invariant structure without asserting normative conclusions as true. Normative reasoning may occur in interpretive layers, where framework assumptions are explicit and disagreement remains representable. Normativity is externalized, not eliminated.

Framework Dependence and Externalization

Framework-Dependent

A property of claims, conclusions, or attributions whose truth or applicability may vary across admissible interpretive frameworks. Causal and normative claims are treated in this paper as framework-dependent under the contestability assumption. Framework-dependent claims may be represented by the substrate, but they may not be asserted as substrate-layer facts when neutrality is required.

Assertion-Level Exclusion

The rule that neutrality restricts what the substrate asserts as true, not which words or predicate symbols may occur in represented content. Causal and normative claims may appear in the substrate as reified, attributed, provenance-bearing assertions. They may not be asserted of substrate individuals as substrate-layer facts.

Representation Without Endorsement

The representation of a claim without asserting that claim as true at the substrate layer. A substrate may record that a framework, agent, document, institution, or record asserted a causal or normative claim. Such representation preserves attribution and provenance while leaving truth-evaluation to the relevant interpretive framework.

Externalized Interpretation

The placement of causal, normative, evaluative, legal, policy-based, or responsibility-bearing conclusions in interpretive layers rather than in the substrate layer. Externalized interpretation allows multiple admissible frameworks to derive incompatible conclusions over shared referents without forcing the substrate to choose among them.

References

- Robert Arp, Barry Smith, and Andrew D. Spear. *Building Ontologies with Basic Formal Ontology*. MIT Press, 2015.
- Paul Schiff Berman. Global legal pluralism. *Southern California Law Review*, 80(6): 1155–1237, 2007.
- Stefano Borgo, Roberta Ferrario, Aldo Gangemi, Nicola Guarino, Claudio Masolo, Daniele Porello, Emanuele M. Sanfilippo, and Laure Vieu. DOLCE: A descriptive ontology for linguistic and cognitive engineering. *Applied Ontology*, 17(1):45–69, 2022a.
- Stefano Borgo, Antony Galton, and Oliver Kutz. Foundational ontologies in action. *Applied Ontology*, 17(1):1–16, 2022b. Editorial introducing special issue comparing BFO, DOLCE, GFO, GUM, TUpper, UFO, and YAMATO.
- Mark Bovens. Analysing and assessing accountability: A conceptual framework. *European Law Journal*, 13(4):447–468, 2007.
- Mark Bovens, Robert E. Goodin, and Thomas Schillemans, editors. *The Oxford Handbook of Public Accountability*. Oxford University Press, Oxford, 2014.
- Geoffrey C. Bowker and Susan Leigh Star. *Sorting Things Out: Classification and Its Consequences*. MIT Press, Cambridge, MA, 1999.
- Joost Breuker and Rinke Hoekstra. Epistemology and ontology in core ontologies: Folaw and Iri-core, two core ontologies for law. *Artificial Intelligence and Law*, 17(3):175–209, 2009.
- Ronald Dworkin. *Law’s Empire*. Harvard University Press, Cambridge, MA, 1986.
- Luciano Floridi. The method of levels of abstraction. *Minds and machines*, 18(3): 303–329, 2008. Defines the epistemic observables available to an agent.
- Luciano Floridi. *The Philosophy of Information*. Oxford University Press, Oxford, 2011. See Chapter 3 for the distinction between LoA (epistemic) and LoO (ontological hierarchy).
- Jonathan Fox. The uncertain relationship between transparency and accountability. *Development in Practice*, 17(4–5):663–671, 2007.

- Thomas R. Gruber. A translation approach to portable ontology specifications. *Knowledge Acquisition*, 5(2):199–220, 1993.
- Nicola Guarino, Daniel Oberle, and Steffen Staab. What is an ontology? In *Handbook on Ontologies*, pages 1–17. Springer, 2009.
- H. L. A. Hart. *The Concept of Law*. Oxford University Press, Oxford, 1961.
- Rinke Hoekstra, Joost Breuker, Marcello Di Bello, and Alexander Boer. The lkif core ontology of basic legal concepts. In *Proceedings of the Workshop on Legal Ontologies and Artificial Intelligence Techniques*, 2007.
- Christopher Hood. *The Blame Game: Spin, Bureaucracy, and Self-Preservation in Government*. Princeton University Press, Princeton, NJ, 2011.
- International Organization for Standardization and International Electrotechnical Commission. Information technology — top-level ontologies (tlo) — part 2: Basic formal ontology (bfo), 2021. Standardizes BFO-2020.
- Marcin P. Joachimiak, Mark A. Miller, J. Harry Caufield, Ryan Ly, Kristofer E. Bouchard, et al. The artificial intelligence ontology: Llm-assisted construction of ai concept hierarchies. *Applied Ontology*, 2025. First published online March 2, 2025.
- Nele Köhler and Fabian Neuhaus. The mercurial top-level ontology of large language models. *Applied Ontology*, 20(1):5–15, 2025. doi: 10.1177/15705838251336685.
- Claudio Masolo, Stefano Borgo, Aldo Gangemi, Nicola Guarino, and Alessandro Oltramari. Wonderweb deliverable d18: Ontology library. *IST Project 2001-33052 WonderWeb*, 2003.
- Claudio Masolo, Francesco Compagno, and Stefano Borgo. On the formal alignment of foundational ontologies: Building mappings between Basic Formal Ontology and Descriptive Ontology for Linguistic and Cognitive Engineering. *Applied Ontology*, 20, 2025.
- Fabian Neuhaus and Janna Hastings. Ontology development is consensus creation, not (merely) representation. *Applied Ontology*, 2022. arXiv:2210.12026.
- Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2 edition, 2009.

- John Rawls. *Political Liberalism*. Columbia University Press, New York, 1993.
- Giovanni Sartor, Pompeu Casanovas, et al. Legislative xml and ontologies in legal drafting. In *Approaches to Legal Ontologies*, pages 239–261. Springer, 2011.
- Barry Smith and Werner Ceusters. Ontology as realist representation. *Applied Ontology*, 5(2):139–188, 2010.
- Peter Spirtes, Clark Glymour, and Richard Scheines. *Causation, Prediction, and Search*. MIT Press, 2nd edition, 2000.
- Susan Leigh Star. The institutional ecology of infrastructure. *Social Studies of Science*, 19(3):327–364, 1989.
- Susan Leigh Star and Karen Ruhleder. Steps toward an ecology of infrastructure. *Information Systems Research*, 7(1):111–134, 1996.
- Langdon Winner. Do artifacts have politics? *Daedalus*, 109(1):121–136, 1980.
- James Woodward. *Making Things Happen: A Theory of Causal Explanation*. Oxford University Press, Oxford, 2003.